Anatomical Structure Segmentation with Partial Annotations: 5th Place Solution for the ENS x Raidium Data Challenge

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Abstract

This report presents the 5th place solution for the ENS x Raidium Data Challenge, focusing on the semantic segmentation of 54 anatomical structures and tumors from 2D CT scans. The primary challenge was the utilization of partially annotated training data. To address this, a custom U-Net architecture was trained from scratch, integrating deep supervision and independent sigmoid activations. A Marginal Dice Loss and Focal Loss were combined to handle missing annotations by leveraging an explicit list of annotated labels. The final model achieved a global Dice score of 0.5569 on the private test set.

Keywords: Semantic Segmentation, Partial Annotations, CT Scan, U-Net, Deep Supervision, Marginal Dice Loss

1 Introduction

Computed Tomography (CT) scans are a primary modality for analyzing human anatomy. Automated segmentation of anatomical structures from these scans is critical for computer-assisted diagnosis. The ENS x Raidium Data Challenge focuses on the semantic segmentation of 54 distinct anatomical structures and tumors, along with a background class, from abdominal CT scans.

A major bottleneck in developing robust medical imaging models is the requirement for exhaustively annotated datasets. In clinical practice, and in this specific challenge, datasets are often compiled from multiple sources with disjoint label sets. This results in partially annotated images where an organ visibly present in the scan may be labeled as background. This discrepancy systematically disrupts the training

of standard semantic segmentation algorithms that rely on mutually exclusive class probabilities.

Furthermore, the challenge imposes strict constraints. The input data consists of randomized 2D grayscale slices (256×256 pixels), removing any 3D spatial context and precluding the use of 3D convolutional architectures. Additionally, the use of external radiological training data or radiologically pre-trained models is strictly prohibited.

To address the partial annotation problem under these constraints, this report details a methodology utilizing a custom U-Net architecture trained entirely from scratch. The primary contributions of this work are:

- The implementation of a 2D segmentation pipeline adapted to isolated slices and highly imbalanced data.
- The design of an objective function combining Marginal Dice Loss and Focal Loss with independent sigmoid activations, leveraging an explicit index of annotated labels to bypass missing annotations.
- An ablation study demonstrating the superiority of a global hard thresholding strategy over class-specific optimized thresholds.

This approach yielded a global Dice score of 0.5569 on the private test set, ranking 5th in the final evaluation. The remainder of this report is organized as follows: Section 2 details the data exploration and preprocessing, Section 3 describes the proposed methodology, and Section 4 presents the experiments, ablation study, and results.

2 Dataset Characteristics and Preprocessing

2.1 Dataset Statistics and Class Imbalance

The training dataset consists of 2000 randomized 2D transverse CT slices (256×256 pixels), split into 800 partially annotated images and 1200 unannotated images. The segmentation task defines 54 distinct anatomical classes plus a background class (value 0).

Given the partial nature of the annotations, the provided `annotated_labels.json` metadata was utilized as a binary filter to determine the absolute presence or absence of specific structures in each slice. This is a crucial step to avoid penalizing the network for predicting anatomically correct but unannotated organs.

An analysis of the dataset characteristics revealed a severe class imbalance. Figure 1 illustrates the log-scale frequency of occurrence for each anatomical structure across the annotated training set.

In addition to this inter-class imbalance (frequency of appearance), there is a significant spatial imbalance. Several small anatomical structures represent a minute fraction of the total pixel volume compared to larger organs (e.g., the liver) or the background. This dual imbalance (frequency and spatial scale) dictates the necessity for specialized objective functions, such as Focal Loss, and specific sampling strategies during the network optimization phase.

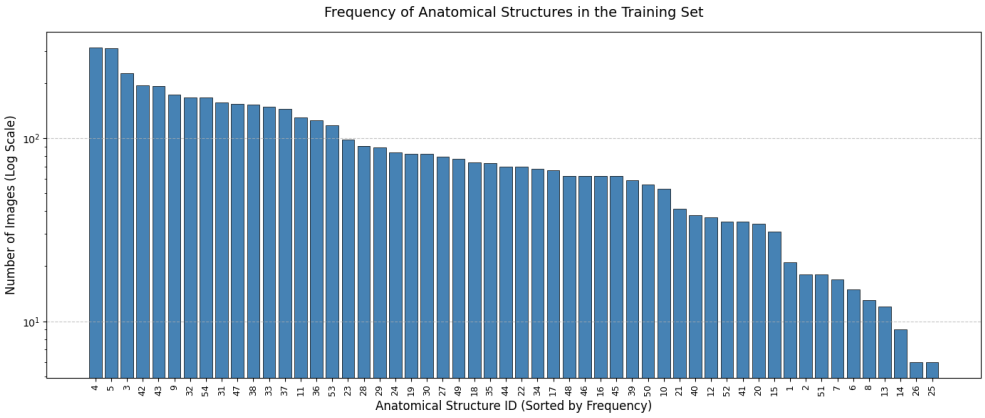

Fig. 1 Log-scale frequency of appearance per anatomical structure across the annotated training set. The long-tail distribution indicates a severe inter-class imbalance.

2.2 Offline Image Preprocessing

The provided CT slices are encoded as 8-bit grayscale images (intensity range). Despite the absence of raw Hounsfield Units (HU), image intensities can still contain residual outlier artifacts that hinder convergence. To stabilize the network optimization, an offline preprocessing pipeline inspired by the nnU-Net heuristic was implemented [1].

First, label masks were extracted from the dense CSV format and converted to individual 2D NumPy arrays (`.npy`) to eliminate I/O bottlenecks during training. Second, global intensity statistics were computed exclusively on the foreground pixels (where mask value > 0) of the 800 annotated slices. The 0.5 and 99.5 percentiles ($p_{0.5}$ and $p_{99.5}$) of these isolated foreground pixels were calculated.

All 2000 images were subsequently clipped to the $[p_{0.5}, p_{99.5}]$ intensity range. Finally, the global mean and standard deviation were computed from these clipped foreground pixels. These global metrics are stored and utilized to apply a standard Z-score normalization to each slice dynamically during the online dataset loading phase.

3 Methodology

To address the challenges of partial annotations and severe class imbalance, we propose a two-phase pipeline. Phase 1 consists of a global 2D U-Net trained with marginal objective functions and an Inverse Square Root (ISQR) sampling strategy. Phase 2 introduces an anchor-based cascade to recover small, underrepresented organs.

3.1 Global Network Architecture and Deep Supervision

The core architecture is a custom 2D U-Net (`PlainConvUNet`) [2] trained from scratch. The network topology is derived from the self-configuring heuristics of nnU-Net [1]. Based on the fixed input patch size of 256×256 pixels, the encoder consists of seven

stages (six pooling operations). This depth ensures a maximum receptive field while maintaining a minimum spatial resolution of 4×4 pixels at the bottleneck, preventing the destruction of spatial representations.

The feature maps scale from 32 channels at the highest resolution to 512 channels at the bottleneck. To mitigate internal covariate shift and stabilize the optimization given the small batch sizes necessitated by the network's depth, each convolutional block utilizes Instance Normalization paired with LeakyReLU activations.

Deep Supervision is integrated into the final three stages of the decoder. During training, the auxiliary outputs from the lower-resolution stages are interpolated to the target resolution. The total loss is a weighted sum of the multi-scale predictions:

$$L_{DS} = \sum_{i=0}^{2} w_i \cdot L(y, \hat{y}_i)$$

where the weights are strictly decreasing ($w = [1.0, 0.5, 0.25]$). During inference, only the final full-resolution layer is utilized.

3.2 Handling Partial Annotations: Independent Sigmoids and Marginal Losses

Standard semantic segmentation frameworks, such as nnU-Net, assume mutually exclusive classes and apply a Softmax activation followed by categorical cross-entropy. In partially annotated datasets, Softmax introduces destructive inter-class competition. If an unannotated organ is present and generates a high logit, the Softmax denominator increases, artificially suppressing the probabilities of surrounding annotated classes, even if the loss for the unannotated class is masked.

To prevent this indirect penalization, the Softmax layer was replaced with C independent Sigmoid activations, transforming the objective into C independent binary classification tasks [3]. Furthermore, we formulated a *Marginal Segmentation Loss* that masks the error computation for unverified pixels using the binary validity vector v derived from the `annotated_labels.json`.

The objective function combines a Marginal Dice Loss (L_{mDice}) and a Marginal Focal Loss (L_{mFocal}):

$$L = 0.6 \cdot L_{mDice} + 0.4 \cdot L_{mFocal}$$

For a batch of images, let V be the subset of classes flagged as "present" or "absent" with certainty by the metadata. The Marginal Dice Loss is computed strictly over this valid subset:

$$L_{mDice} = 1 - \frac{1}{|V|} \sum_{c \in V} \frac{2 \sum_i (\hat{p}_{i,c} \cdot y_{i,c}) + \epsilon}{\sum_i \hat{p}_{i,c} + \sum_i y_{i,c} + \epsilon}$$

where $\hat{p}_{i,c}$ is the sigmoid probability for pixel i and class c, and $y_{i,c}$ is the binary ground truth. The Marginal Focal Loss applies a dynamic scaling factor $(1 - \hat{p}_t)^\gamma$ exclusively to the valid pixels to manage extreme background dominance [4].

3.3 Inverse Square Root (ISQR) Sampling

An Inverse Square Root (ISQR) weighted random sampler [5] was employed during batch generation. The sampling weight W_c for a class c is inversely proportional to the square root of its frequency:

$$W_c = \min\left(\frac{1}{\sqrt{f_c/N}}, W_{max}\right)$$

where f_c is the number of images containing class c, N is the total number of annotated images, and $W_{max} = 15$ acts as a clipping threshold. Each image is assigned the weight of its rarest annotated organ.

3.4 Anchor-Based Cascade for Small Structures

Small anatomical structures (labels L1 and L2) exhibited near-zero performance in the global model due to their minute spatial footprint. Note that all label identifiers (L1, L2, L3, L33) follow the anonymized class numbering of the ENS × Raidium challenge; semantic mappings were not disclosed to participants. Inspired by coarse-to-fine cropping methodologies [6], we developed a secondary anchor-based cascade.

The trained global model generates "fused masks" by inferring pseudo-labels for unannotated pixels with high confidence ($p > 0.5$). An offline spatial analysis on these masks determined stable geometric relationships (center-of-mass distance and directional vectors) between small target organs and larger stable organs.

The analysis identified organ L3 as a highly reliable anchor for L1, present in 100% of co-occurring cases with a mean distance of 19.9 ± 3.8 pixels and a constant directional vector. Similarly, L33 was identified as the optimal anchor for L2. Figure 2 illustrates these spatial relationships across multiple scans.

During cascade training, a secondary `ParametricUNet` operates strictly as a binary segmenter. Its input is a tightly cropped patch dynamically extracted around the center-of-mass of the predicted anchor organ, offset by the pre-computed spatial vector. This isolates small structures from the global background, drastically simplifying optimization.

4 Experiments and Results

4.1 Experimental Setup

4.1.1 Validation Strategy

Given the severe class imbalance and the rarity of certain structures (e.g., some organs appearing in only 6 images), a standard random split would fail to evaluate the model correctly. We implemented a stratified validation split to guarantee that every single anatomical structure (all 54 classes) is represented in the validation set.

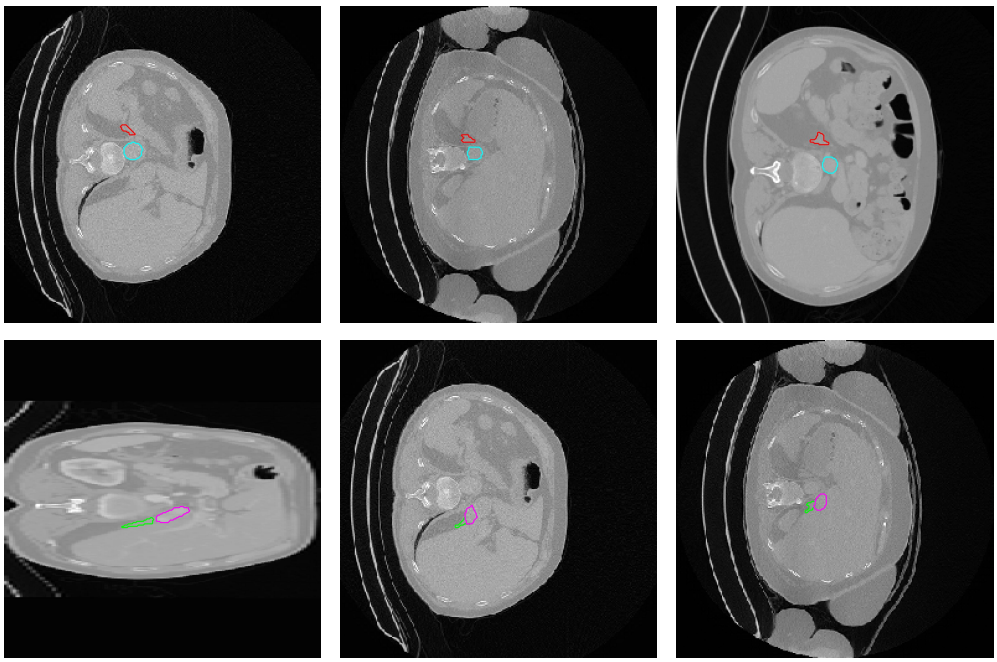

Fig. 2 Spatial relationships between target organs and their respective anchors. Top row: Target L1 (Red) relative to Anchor L3 (Cyan). Bottom row: Target L2 (Lime) relative to Anchor L33 (Magenta). These stable geometric priors dictate the bounding box definitions for the secondary cascade network extraction.

4.1.2 Data Augmentation

To prevent overfitting on the limited set of 800 annotated images, a robust on-the-fly data augmentation pipeline was applied, heavily inspired by the nnU-Net heuristics for medical imaging [1]. Spatial transformations included random affine scaling (0.85 to 1.25), translation, rotation ($\pm 15°$), and elastic deformations ($\alpha = 100, \sigma = 5$) to simulate natural anatomical variations. Intensity transformations included Gaussian noise, Gaussian blur, random brightness, and gamma corrections.

4.1.3 Training Hyperparameters

The global model (Phase 1) was optimized using Stochastic Gradient Descent (SGD) with Nesterov momentum (0.99) and a weight decay of 3×10^{-5}. We utilized a large batch size of 42, derived from the nnU-Net self-configuring heuristic based on the fixed patch size of 256×256 and available GPU memory. The learning rate was initialized at 0.01 and decayed using a Polynomial Learning Rate Scheduler (power of 0.9) over 500 epochs, with each epoch defined as 250 iterations (125,000 total iterations).

The secondary cascade network (Phase 2) was trained on localized patches with a batch size of 4. We employed the AdamW optimizer with a learning rate of 2×10^{-4} and a Cosine Annealing scheduler over 300 epochs. Spatial augmentations were constrained to minimal translations to preserve the anchor-based bounding box integrity.

4.2 Ablation Study and Results

We conducted an ablation study to evaluate the impact of our architectural modifications and post-processing strategies. The primary metric is the macro-averaged Dice score across all classes and images on the private test set. Results are summarized in Table 1.

Table 1 Ablation study on the private test set. The baseline nnU-Net fails due to mutually exclusive Softmax activations penalizing partial labels. DS = Deep Supervision.

Model Architecture	Loss / Post-processing	Global Dice Score
nnU-Net (Baseline)	Softmax (CE + Dice)	0.2905
Custom U-Net (w/o DS)	Marginal Loss + Class-Specific Thresholds	0.5188
Custom U-Net (with DS)	Marginal Loss + Class-Specific Thresholds	0.5403
Custom U-Net (with DS)	Marginal Loss + Global Threshold (0.5)	0.5559
Custom U-Net + Cascade	Marginal Loss + Global Threshold (0.5)	**0.5569**

Baseline Failure: The standard nnU-Net baseline, relying on Softmax and Cross-Entropy, achieved a low score of 0.2905. This empirically confirms the hypothesis that forcing mutual exclusivity on partially annotated data degrades representation learning.

Deep Supervision: Integrating Deep Supervision to the Custom U-Net improved the score from 0.5188 to 0.5403 (+0.0215). This demonstrates improved gradient flow and feature extraction in the intermediate decoder layers.

Thresholding Strategy: We initially hypothesized that optimizing the binarization threshold independently for each class on the validation set would yield better performance. However, evaluating this strategy against a strict global threshold of 0.5 across all classes revealed that class-specific thresholds overfit the validation set. The global threshold generalized better, increasing the score to 0.5559.

Cascade Impact: The implementation of the Phase 2 anchor-based cascade for classes L1 and L2 provided a final performance gain, pushing the global Dice score to 0.5569, establishing our final ranking.

5 Conclusion

In this report, we presented a robust methodology for the semantic segmentation of 54 anatomical structures from 2D CT slices under severe partial annotation constraints. By substituting the standard Softmax formulation with independent sigmoid activations and a Marginal Segmentation Loss guided by metadata, we successfully prevented the network from penalizing unannotated ground truth structures. Furthermore, the combination of Inverse Square Root sampling and a secondary anchor-based cascade network effectively addressed both frequency and spatial class imbalances. This two-phase pipeline achieved a global Dice score of 0.5569, demonstrating high efficacy in extracting geometric and semantic features from incomplete medical datasets.

Code Availability. The complete source code for data preprocessing, model training, and inference, including the implementation of the Inverse Square Root (ISQR) sampling and the anchor-based cascade, is publicly available on GitHub. The repository can be accessed at: https://github.com/bryan29-ly/ENSxRaidium-Data-Challenge

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References

[1] Isensee, F., Jaeger, P.F., Kohl, S.A.A., Petersen, J., Maier-Hein, K.H.: nnu-net: a self-configuring method for deep learning-based biomedical image segmentation. Nature methods **18**(2), 203–211 (2021)

[2] Ronneberger, O., Fischer, P., Brox, T.: U-net: Convolutional networks for biomedical image segmentation. In: Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015, pp. 234–241 (2015). Springer

[3] Shi, G., Xiao, L., Chen, Y., Zhou, S.K.: Marginal loss and exclusion loss for partially supervised multi-organ segmentation. Medical Image Analysis **71**, 102046 (2021)

[4] Lin, T.-Y., Goyal, P., Girshick, R., He, K., Dollár, P.: Focal loss for dense object detection. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 2980–2988 (2017)

[5] Mahajan, D., Girshick, R., Ramanathan, V., He, K., Paluri, M., Li, Y., Bharambe, A., Maaten, L.: Exploring the limits of weakly supervised pretraining. In: Proceedings of the European Conference on Computer Vision (ECCV), pp. 181–196 (2018)

[6] Wasserthal, J., Breit, H.-C., Meyer, M.T., Pradella, M., Hinck, D., Sauter, A.W., Heye, T., Boll, D.T., Cyriac, J., Yang, S., et al.: Totalsegmentator: Robust segmentation of 104 anatomic structures in ct images. Radiology: Artificial Intelligence **5**(5) (2023)